THYROID DRUGS

BIETE LUNDAU LUKE

DipPharm, BPharm, MclinPharm

OVERVIEW

- The thyroid gland synthesizes and secretes **Triiodothyronine (T3)** which is the more active and **Tetraiodothyronine (T4)** which is also known **as thyroxine**

- The thyroid hormones are necessary for normal growth, development and timely sexual maturation

- They affect every system and have a crucial role in metabolic processes including the synthesis and degradation of all other hormones

Overview Cont'd

- Thyroid hormones also augument sympathetic nervous system function by primarily increasing the number of adrenoceptors in target tissues

- Thyroid hormones are synthesized in a process that involves the uptake and organification of iodide and the subsequent coupling of iodotyrosine residues of thyrogloburin

- Secretion of thyroid hormones is initiated by a hypothalamic hormone called **thyrotropin releasing hormone (TRH)** which increases the secretion of the anterior pituitary hormone called **thyroid stimulating hormone (TSH or thyrotropin)**

Summary of Thyroid disorders

- Normal thyroid function (Euthyroidism) is maintained via feedback inhibition of TSH secretion

- This helps to keep the plasma concentration of free (Circulating or unbound) T4 within a narrow acceptable range

- Abnormally low or high T4 and T3 levels results in clinical manifestations of hypothyroidism and hyperthyroidism respectively

- Hypothyroidism is characterized by low T4 levels which leads to impaired growth and development and also decreased metabolic activity

- Hyperthyroidism on the other hand is characterized by high T4 levels which leads to hyperactivity of systems (particularly the cardiovascular and nervous system) and increased metabolic activity

Diagnosis of hypothyroidism

- Thyroid disorders are diagnosed on the basis of their clinical manifestation and plasma T4 and TSH levels

- In most cases, TSH plasma levels are abnormally high in hypothyroidism and low in hyperthyroidism

Symptoms of thyroid diseases

Symptoms of Hypothyroidism

- Bradycardia (heart rate)
- Fatigue
- Unexplained weight gain
- Dry skin
- Dry and coarse hair
- Depressed mood
- Menorrhagia (Heavy and prolonged menses)

Symptoms of Hyperthyroidism

- Tachycardia
- Difficulty sleeping
- Unexplained weight loss
- Feeling sensitive to heat
- Clammy or sweaty skin
- Anxiety and high irritability
- Irregular menses or amenorrhea (Absence of menses)

DRUGS USED IN HYPOTHYROIDISM

Thyroid hormone preparations

- Levothyroxine (T4), liothyronine (T3), Liotrix (Combination of T4and T3 in the ratio of 4:1)
- These are synthetic drugs used for thyroid replacement therapy in patients with hypothyroidism
- Thyroid dessicated is a natural preparation of thyroid gland from pigs which contains both T4 and T3
- Controlled trials showed that the combination of T4 and T3 was not superior to T4 alone with respect to fatigue, depression, quality of life or other symptoms associated to hypothyroidism
- Thus liotrix and thyroid dessicated are not recommended by many endocrinologists for the management of hypothyroidism

Levothyroxine

- Oral bioavailability is about 80%
- Diffferent brands and generic formulations of levothyroxine vary in hormonal content and bioavailability
- Thus formulations should not be substituted for one another without monitoring T4 and TSH levels
- Food also affects the bioavailability of levothyroxine and hence recommended that it be taken at the same mealtime every day
- Because of having a half life of 7 days, once daily administration of the drug produces little fluctuations in plasma

Indications of levothyroxine

- Levothyroxine is the drug of choice for thyroid hormone replacement in patients with hypothyroidism

- This is because it is chemically stable, non allergic and can be given once daily

- Levothyroxine produces a stable pool of T4 that is converted to T3 at a steady and consistent rate

- Drug is started at a lower dose which is the then increased at a monthly interval until a full replacement dose is achievement

Adverse effects of levothyroxine

- Thyroid hormone preparations rarely cause adverse reactions if dose is appropriate and monitored during initiation of treatment and periodically thereafter

- Excessive doses produce symptoms of hyperthyroidism

Interactions

- Aluminium hydroxide, calcium supplements, cholestyramine, ferrous sulphate and sucrafate are among the drugs that interfere with levothyroxine absorption

- Therefore, if levothyroxine is to be prescribed to a patient also taking the above mentioned drugs, they should be given 2 hours apart

- Estrogen, androgens and glucorticoids can alter thyroid binding globulin and T4 and T3 levels but free T4 and TSH remain normal in patients taking these steroids

Liothyronine

- Liothyronine (T3) is more potent than levothyroxine(T4) and has a higher bioavailability
- It is however, seldomly used in treatment of hypothyroidism because it causes more adverse cardiac effects and is more expensive than levothyroxine
- Has a shorter life half life than levothyroxine leading to the need for multiple doses to achieve required response
- Liothyronine does not increase T4 levels and so its difficult to monitor the response to treatment

DRUGS USED IN HYPERTHYROIDISM

Introduction of Hyperthyroidism

- Graves disease, an autoimmune disease that affects the thyroid, is the most common cause of hyperthyroidism

- In these situations, TSH levels are reduced due to negative feedback due to high levels of circulating thyroid hormone

Goal of therapy

- Decrease synthesis of the thyroid hormone
- Decrease release of the additional thyroid hormone

Modalities of accomplishing treatment goals

- This can be accomplished by;

1. Removing part or all of the thyroid gland

- surgically or by destruction of the gland with radioactive iodine (I-131), which is selectively taken up by the thyroid follicular cells

2. Inhibiting synthesis of the hormones

- Thiomides e.g. Methimazole and propyltiouracil

  NB: Carbimazole is a pro drug of methimazole

3. Blocking release of the hormones from the follicle

- Iodide salts

Anti-thyroid Agents

Anti-thyroid agents used in the treatment of hyperthyroidism include;

- **Thiomide drugs**
  - Methimazole
  - Propylthiouracil (PTU)
- Beta adrenoceptor antagonists e.g. propranolol
- Iodide salts
- Radioactive isotope (RAI)

Thiomide drugs

- Methimazole and propylthiouracil

Mechanism of action

- Thyroid hormone requires oxidation of trapped iodide, formation of iodotyrosines and the coupling of iodotyrosines to form T3 and T4

- Methimazole and PTU inhibit thyroperoxidase catalyzed steps in this process

- Additionally, PTU inhibits the conversion of T4 to T3 though it is evidenced that methimazole and PTU appear to be therapeutically equivalent

Pharmacokinetic

- Thiomides are well absorbed from the gut after oral administration

- They are actively concentrated in the thyroid gland which may account for their relatively longer duration of action despite having a short half

- Thiomides are extensively metabolized before undergoing renal excretions

Indications of thiomides

- In Graves disease, thiomides are used in an attempt to induce remission or a means to control symptoms before surgery or RAI treatment

- Effects of thiomides are delayed because it takes about 4 to 8 weeks of therapy before the glandular hormone is depleted and the circulating hormone begins to go to normal

Adverse effects

- Pruritic maculopular rash
- Athralgia
- Fever
- Methimazole is more likely to cause birth defects and hence PTU is the drug of choice just before and during first trimester of pregnancy
- Less frequently occurring effects are hepatitis, gastrointestinal distress

Beta adrenoceptor antagonists

- Thyroid hormones and the sympathetic nervous system act synergistically on the cardiovascular system
- Increased levels of thyroid hormones therefore, causes tachycardia, palpitation and arrhythmias
- Beta adrenoceptor antagonists are used to reduce cardiovascular stimulation associated with hyperthyroidism
- They act immediately and particularly useful during severe thyrotoxicosis also known as **thyroid storm**
- They are also used to control symptoms of hyperthyroidism in patients awaiting either surgery or a response to RAI treatment

Other anti-thyroid drugs

Iodide salts

- Iodide salts are contained in potassium iodide tablets and solutions such as saturated solution of potassium iodide and Lugol solution (elemental iodine and potassium iodide)

Mechanism of Action of Iodide Salts

- Iodide causes a transient decrease in thyroid hormone synthesis

- Excess iodine inhibits the function of thyroperoxidase by reducing iodide oxidation and organification which results in an inhibition of the release of the thyroid hormones from the gland

- This mechanism is called **Wolff-Chaikoff** effect which is an autoregulatory mechanism of the thyroid gland to handle excess iodine intake and thus subsequently preventing thyroid hormone formation

Indications of Iodide salts

- Potassium iodide is used in acute thyrotoxicosis to prepare patients with acute thyrotoxicosis both to prepare patients for surgery or to inhibit the release of thyroid hormones after RAI treatment
- When administered in sufficient doses, iodide salts immediately act to inhibit the release of thyroid hormones from the gland
- In patients scheduled for thyroid surgery, a potassium iodide solution is usually administered preoperatively 7 to 14 days to reduce the size and vascularity of the gland

Side effects of iodides

- Usually mild and can include skin rashes and other hypersensitivity reactions
- Salivary gland swelling
- Metallic taste
- Sore gums
- Gastrointestinal discomfort

Radioactive iodide (RAI)

- Usually administered as a colorless and tasteless solution of sodium iodide (I-131) and it is quickly absorbed from the gut and then concentrates in the thyroid
- In the thyroid, it emits beta particles which destroy thyroid tissue which leads to the gradual normalization of the circulating thyroid hormone
- Beta adrenoceptor antagonists are used to control symptoms of hyperthyroidism while the patient is awaiting the response to RAI treatment

Radioactive iodide (RAI) Cont'd

- Methimazole or PTU can be used if beta blockers are not adequate to control these symptoms however, thiomide drugs should be withdrawn several days before initiating RAI so as to optimize the cure rate and prevent the recurrences

- Iodide salts are used after RAI treatment to inhibit radioactive thyroid but should not be used before RAI because nonradioactive iodide compete with radioactive iodide for uptake by the thyroid gland

- RAI is absolutely contraindicated in pregnant women because it destroys fetal thyroid tissue

END